

Private Team Chat Management System

Client Workflow & Live Demonstration Guide

Product Type:	Multi-Tenant Team Messaging SaaS
Target Users:	Enterprises, Regulated Teams, Compliance
UI/UX Paradigm:	WhatsApp Web & Mobile App Experience
Security Architecture:	PostgreSQL RLS + PII Sanitization
Realtime Latency:	< 50ms (Zero-Refresh Event Sync)
Prepared Date:	September 2026

Purpose of this Document: This document outlines the end-to-end architecture, user hierarchy, security guardrails, and step-by-step walkthrough scripts for evaluating the Private Team Chat Management System.

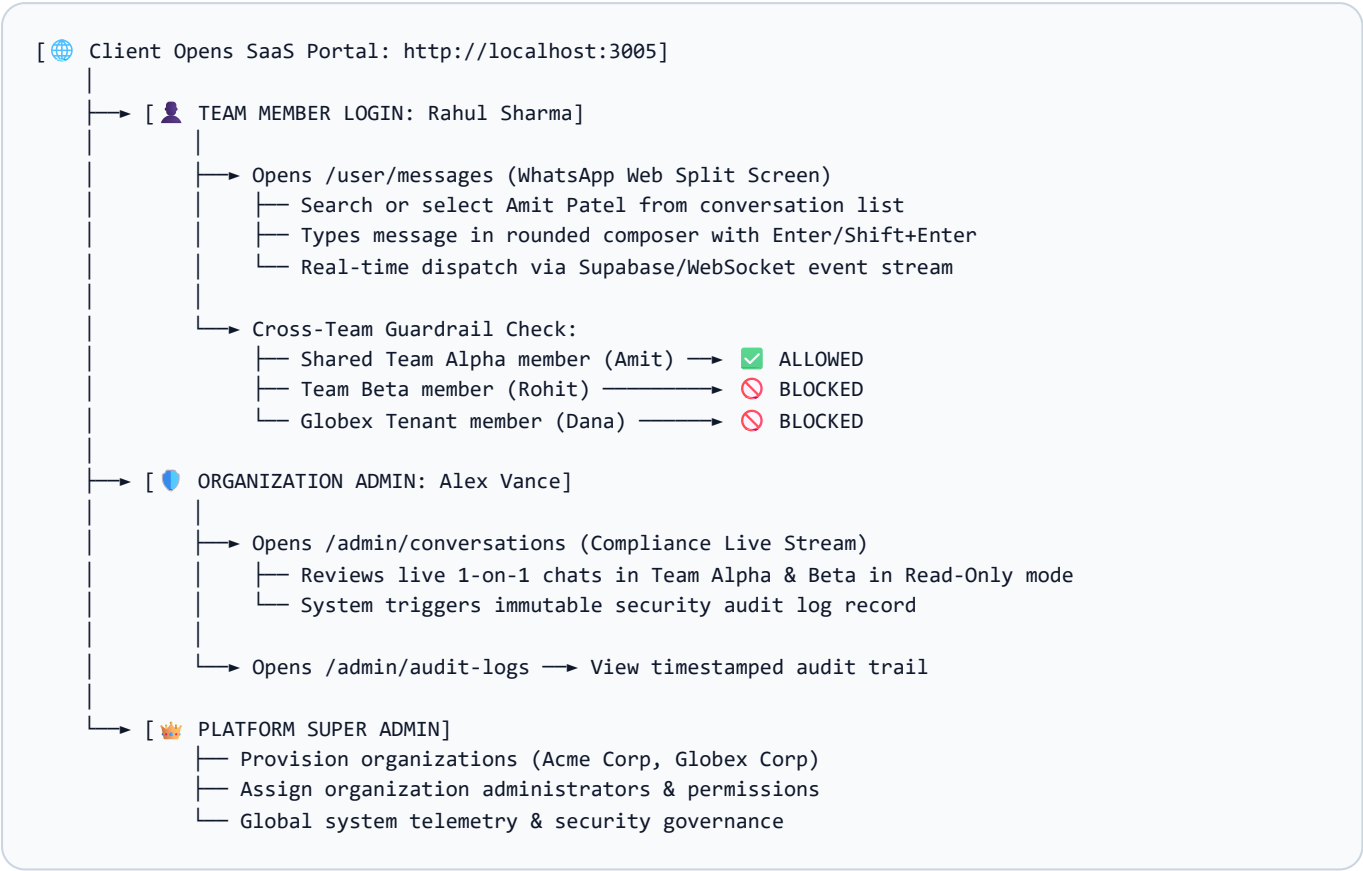
1. Organization & Role Hierarchy (Tree Structure)

The system enforces a strict multi-level governance structure. Communication and administration are strictly bound by tenant and team membership:



2. End-to-End User Action & Message Flow

Below is the decision-tree and sequence followed when team members and administrators interact with the platform:



3. Roles, Scopes & Permissions Matrix

Role	Authorized Scope	Chat Capabilities	Admin Oversight	PII Visibility
SUPER_ADMIN	Global Platform	Global conversation inspector	Full cross-org access	Full Administrative
ORG_ADMIN	Assigned Teams in Org	Read-only compliance audit	Assigned teams only	Org Member Emails
USER	Shared Teams Only	1-on-1 WhatsApp-style chat	Zero access (Strict Deny)	Sanitized (@username only)

4. 5-Minute Live Client Walkthrough Script

Use this exact 10-step sequence when conducting a live demonstration to prospective clients or stakeholders:

Step	Demo Action	What to Say to the Client	Expected Result
1	Open Landing Page	"This is our SaaS portal displaying multi-tenant governance with instant 1-click evaluation personas."	http://localhost:3005
2	Select Rahul	"We are logged in as Rahul Sharma, an engineer in Team Alpha. Notice that he only sees his team's environment."	User Dashboard
3	Open Messages	"Here is the modern WhatsApp Web chat UI: circular avatars, search filter, green unread badges, and textured wallpaper."	/user/messages
4	Launch 2-User Demo	"Let's demonstrate instant two-way real-time messaging between two simultaneous users (Rahul and Amit)."	/demo/chat
5	Send Message	"When Rahul sends a message, watch Amit's screen on the right — it delivers instantly without page refresh."	Instant Delivery
6	Reply from Amit	"When Amit replies, Rahul receives it immediately, and message delivery checkmarks turn to blue read receipts."	Double Blue Checks
7	Cross-Team Test	"Rahul cannot chat with Rohit because Rohit belongs to Team Beta. Cross-team isolation is enforced at the database level."	Boundary Isolation
8	Switch to Admin 1	"Now let's switch to Alex Vance (Admin 1). Admins can inspect live team chats in read-only compliance mode."	/admin/conversations
9	Inspect Audit Log	"Every admin inspection automatically generates an immutable audit record for compliance and legal accountability."	/admin/audit-logs
10	Switch to Super Admin	"Super Admins govern multiple corporate client organizations, manage system admins, and monitor global activity."	/super-admin

5. Enterprise Compliance & Security Highlights

 **Multi-Tenant RLS Architecture**

PostgreSQL Row Level Security ensures tenant data is completely isolated. Organization A cannot access Organization B records.

 **Zero PII Exposure (Privacy Guard)**

Regular team members cannot see email addresses or phone numbers. Only public display names and usernames are exposed.

 **Tamper-Proof Audit Logging**

All administrative oversight actions append timestamped records into the audit database, preventing unauthorized snooping.

 **Full Mobile & Desktop Support**

Adaptive responsive UI: WhatsApp Web on desktop/tablets and WhatsApp Mobile App single-view navigation on phones.